

Reconstruction of electroencephalographic data using radial basis functions



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HIGHLIGHTS

- A new interpolation method for EEG reconstruction is introduced.
- The predictive value of the new technique is superior to some commonly used interpolation methods.
- 40 samples from 10 subjects were investigated by cross-validation.

ABSTRACT

Objective: In this paper we introduce a new interpolation method to use for scalp potential interpolation. The predictive value of this new interpolation technique (the multiquadric method) is compared to commonly used interpolation techniques like nearest-neighbour averaging and spherical splines.

Methods: The method of comparison is cross-validation, where the data of one or two electrodes is predicted by the rest of the data. The difference between the predicted and the measured data is used to determine two error measures. One is the maximal error in one interpolation technique and the other is the mean square error. The methods are tested on data stemming from 30 channel EEG of 10 healthy volunteers.

Results: The multiquadric interpolation methods performed best regarding both error measures and have been easier to calculate than spherical splines.

Conclusion: Multiquadrics are a good alternative to commonly used EEG reconstruction methods.

Significance: Multiquadrics have been widely used in reconstruction on sphere-like surfaces, but until now, the advantages have not been investigated in EEG reconstruction.

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1. Introduction

The reconstruction of data by interpolation methods is of great practical interest in EEG research as it is a basis for brain-mapping of multichannel data. In addition, interpolation algorithms can be used for the reconstruction of missing data, which were lost due to technical problems (e.g., caused by a broken electrode or by technical or physiological artefacts). Methods stemming from approximation theory have been applied to this problem, yet several difficulties remain to this date. There are problems computing

the reconstruction, and with most methods used, there is an error increase when the corner electrodes are to be reconstructed. In the present contribution we investigate whether the application of radial basis functions has advantages as compared to the commonly used nearest-neighbour averaging and spherical-splines methods. In addition, we are interested in whether the radial basis approximants are easier and faster to compute. Radial basis functions are used in many different areas, like the reconstruction of temperature data in the oceans (Buhmann, 2000, Section 6.2) or modelling neural networks (Aslan et al., 2008; Chen et al., 1991), and good results have been achieved especially when the approximation of sphere-like surfaces was investigated (Buhmann, 2003, Chapter 1). In particular, a spherical model is the simplest one resembling the real anatomical setting.

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A radial basis function is a real-valued function whose value depends solely on its radial distance to the origin. For interpolation, the origin is translated to a given data point, so that the value of the respective radial basis function depends only on the distance between a given data point and the evaluation point. A weighted sum of those radial basis functions is then used to approximate the unknown data. A significant advantage of using certain radial basis functions is the fact that an interpolation is always possible and independent of the number of data points and their distribution (Buhmann, 2003, Chapter 2.1).

In this study we will compare the commonly used interpolation techniques in EEG reconstruction to the multiquadric techniques, which to our knowledge have not been investigated in this area until now. Yao (2002) described the advantages of radial basis function for the computation of the scalp Laplacian, but focused on a different basis function, namely the Gaussian basis function. His results were questioned by Bortel and Sovka (2013). Both studies mainly compared the scalp Laplacian and only used the Gaussian radial basis function. We will compare the techniques on the level of potential reconstruction only.

We will show that radial basis functions are well-suited as a method for the interpolation of EEG data, and will also try to investigate if using Euclidean or geodesic distances yields better results. To achieve this, we will test one commonly used local technique, the nearest-neighbour technique, two Euclidean techniques (the new multiquadric and thin-plate-splines) and two spherical techniques, that is, techniques using geodesic distances (spherical multiquadrics and commonly used spherical splines).

We used multichannel EEG data from 10 healthy volunteers recorded in four recording situations. For a comparison of the algorithms' performance we will show how the interpolation works when data of one or two electrodes are missing.

2. Methods

2.1. Interpolation by nearest-neighbour-averaging

The nearest-neighbour technique (NN) has been widely used for reconstruction and interpolation of EEG data. The estimated value at a point p by the k -th nearest-neighbour method is given by:

$$v(p) = \text{mean}(\{v(p_1), \dots, v(p_k)\}),$$

where $p_1, \dots, p_k$ denote the positions of the k nearest neighbour electrodes and $v(p_i)$ is the potential measured at this point. The neighbours were determined from the arrangement shown in Fig. 1. In our study we use a maximum of $k = 4$ neighbouring electrodes. We did not weigh the electrodes by their distance from the electrode to be reconstructed, because they are nearly uniformly distributed. The 30 electrodes were regularly spaced starting at theinion, and extended in steps of 15% of the nasion-inion distance to 5% anterior of F_z . The distribution is shown in 1.

The minimal number of neighbours possible is therefore 2.

2.2. Radial basis functions

The radial basis function method requires using the electrode positions in 3-dimensional Euclidean space. We denote $p_i = (x_i, y_i, z_i)$ as the position of the i -th electrode. We chose the 2-sphere of radius one as a model for the scalp, the electrode positions are therefore given as distributed on S^2 . This choice has the advantage that the information accompanying the EEG-Mask used, often include these kind of coordinate-distributions. The interpolant has the general form:

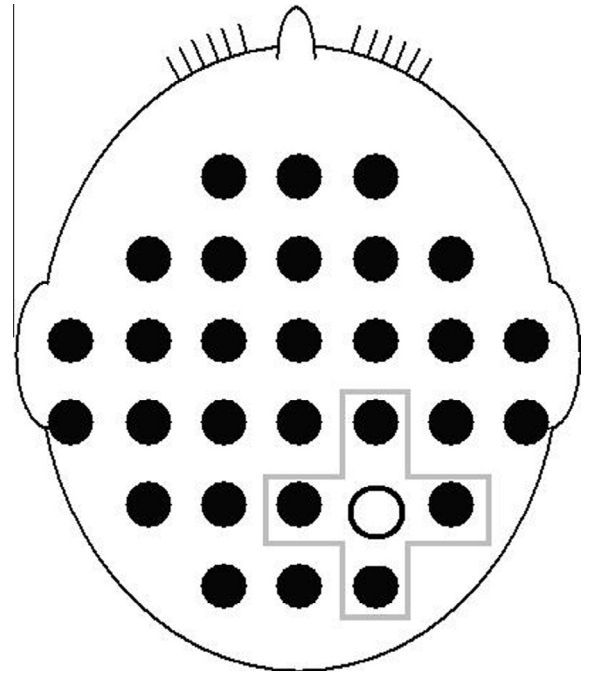


Fig. 1. Distribution of the electrodes on the head, as seen from above.

$$s(p) = \sum_{j=1}^n b_j \cdot \phi(\|p - p_j\|_2),$$

where n is the number of electrodes (not including the ones that have to be reconstructed), $b_j \in \mathbb{R}$ are real coefficients and $\phi: \mathbb{R}^+ \rightarrow \mathbb{R}$ is a radial basis function.

The interpolant is computed by solving the linear system

$$v = A_{\Xi} b,$$

where $A_{\Xi} = \{\phi(\|p_i - p_j\|_2)\}_{i,j=1}^n$, is the so-called interpolation matrix, $v = \{v(p_i)\}_{i=1}^n$ is the vector of data at the respective sites (in this case the potential measured at the electrode positions) and $b = (b_1, \dots, b_n)^T$ is the vector of coefficients which has to be computed. The problem is solvable for scattered data by using positive definite or conditionally positive definite basis functions (Buhmann, 2003) like:

- Thin-plate-splines: $\phi(r) = r^2 \log(r)$,
- Multiquadrics: $\phi(r) = \sqrt{r^2 + c^2}$, $c \in \mathbb{R}$,
- Inverse multiquadrics: $\phi(r) = (r^2 + c^2)^{-\frac{1}{2}}$, $c \in \mathbb{R} \setminus \{0\}$,
- Gaussians: $\phi(r) = e^{-\alpha r^2}$, $\alpha > 0$.

2.2.1. Multiquadrics

One of the methods investigated was the interpolation with multiquadrics (MQ). The basis function used in this context is therefore described by

$$\phi(r) = \sqrt{r^2 + c^2}, \quad r \in \mathbb{R}^+, c \in \mathbb{R}, \quad (1)$$

where c is a parameter to be determined by cross-validation.

We tested 101 different values for c between 0 and 1. We used a subset of 10 sets of EEG data stemming from 10 different volunteers, each including 2501 consecutive points in time. The errors were computed using leave-one-out cross-validation.

Algorithm 1. Leave-one-out-crossvalidation.

Divide the data in n parts $T_1, \dots, T_n$, where T_i denotes the data measured at the i -th electrode.

FOR $i = \{1, \dots, m\}$ DO

1. Use $\{T_1, \dots, T_n\} \setminus T_i$ as a *training set*, and compute the reconstruction $\tilde{T}_i$ of the *test set* T_i .
2. Compare the reconstruction $\tilde{T}_i$ to the original test set T_i by calculating
 - (a) $\text{mean}(T_i - \tilde{T}_i)$ and
 - (b) $\text{var}(T_i - \tilde{T}_i)$.

The mean of the results is shown in Fig. 2. All error graphs for the datasets had a similar form. We therefore decided to use the value $c = 0.07$ for the rest of the investigation.

2.2.2. Thin-plate-splines

The second radial basis function investigated is a modified thin-plate-spline (TPS) (Hubbert and Morton, 2004). The thin-plate-spline interpolant is a function passing through the given points minimising the so-called “bending-energy” (Duchon, 1977). The radial basis function is given by

$$\phi(r) = r^2 \log(r).$$

2.3. Spherical radial basis functions

In this paper, radial basis functions designed for the sphere were of special interest, because we examine a projection of the electrode positions on the 2-sphere. The interpolant then has the form

$$s(p) = \sum_{i=1}^n b_i \phi(d(p, p_i)),$$

where $\phi: [0, \pi] \rightarrow \mathbb{R}$ denotes the basis function and $d(p, p_i)$ is the geodetic distance between p and p_i (Fasshauer and Schumaker, 1998). The geodetic distance measures the smallest distance between two points measured on the surface of the sphere.

2.3.1. The spherical multiquadric

The spherical multiquadric (SM) is a spherical basis function for which solvability of the interpolation problem has been shown for arbitrary point distributions and numbers (Hardy, 1990). It is given by

$$\varphi(t) = \sqrt{1 + \gamma^2 - 2\gamma \cos(t)}, \quad \gamma > 0.$$

For the choice of $\gamma = 1$, the spherical multiquadric is the same as the Euclidean multiquadric in $\mathbb{R}^3$ with $c = 0$. To select a good value for the parameter γ , we used a leave-one-out cross-validation. We used the same subset of 10 datasets stemming from 10 different subjects that we used to determine the smoothing parameter c for the multiquadric interpolation. 201 values of γ between 0 and 2 were tested, and the mean of the errors computed. The results are shown in Fig. 3. The symmetry in the error graph is due to the term γ appearing twice in the function $\varphi(t)$, once with a positive and once with a negative sign. For values just a bit below or above one, there is a compensation between the positive and negative sign part, and the functions $\varphi(t)$ behave very similar. The best result was given using $\gamma = 1.07$, so we decided to choose $\gamma = 1.07$ for further investigation.

2.3.2. Spherical splines

An interpolation method for the sphere, first described by Grace Wahba, is the interpolation with spherical splines (SS) (see: Wahba, 1981).

The corresponding radial basis function has the form

$$\varphi(t) = \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{2n+1}{n^m(n+1)^m} P_n(\cos(t)),$$

where $P_n(x)$ is the n -th Legendre polynomial defined as

$$P_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k \frac{(2n-2k)!}{(n-k)!(n-2k)!k!2^n} x^{n-2k}.$$

Here $\lfloor x \rfloor$ denotes the largest integer less than $x \in \mathbb{R}$. This spherical radial basis function is positive definite (Schoenberg, 1942; Wahba, 1981), which means that the interpolation problem is always solvable. Here, m is a parameter to be chosen from $\mathbb{N}$. To compute the spherical spline interpolant, we used a program by Weber (2005), which we adapted to our problem and which implements the method described by Perrin et al. (1989). A few initial tests showed that the so called cosine matrix.

$$\text{COS} = \{\cos(d(p_i, p_j))\}_{i,j=1}^n,$$

was ill-conditioned with a numerically calculated condition-number of $\text{cond} = 4.38 \cdot 10^{19}$.

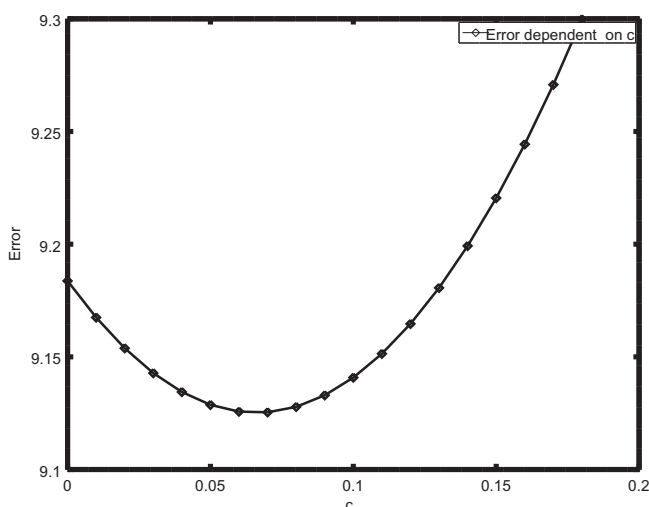


Fig. 2. The error using multiquadric reconstruction dependent on the smoothing-parameter c .

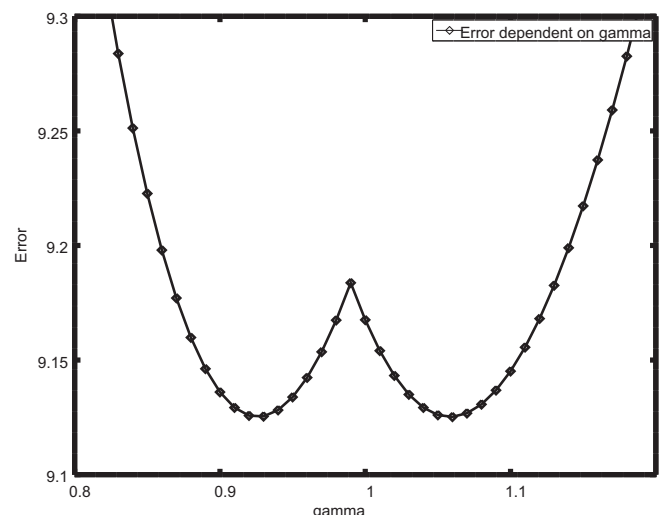


Fig. 3. The error function using spherical-multiquadric reconstruction dependent on the smoothing parameter γ .

It is furthermore not possible to calculate the infinite sum

$$\varphi(t) = \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{2n+1}{n^m(n+1)^m} P_n(\cos(t))$$

needed for the computation of the interpolant, and the sum is therefore approximated by the finite sum

$$f_{N,m}(x) = \frac{1}{\pi} \sum_{n=1}^N \frac{2n+1}{n^m(n+1)^m} P_n(x).$$

Ferree suggests to choose $m \geq 2$ and notes that the use of the first 7 terms of the sum is sufficient in this case. Most studies in the past used $m = 3$ or $m = 4$ (Perrin et al., 1989; Soong et al., 1993).

It is also described by Ferree (2006) that for some $m \geq 3$ (including $m = 3$ and $m = 4$) the infinite sum can be approximated by $g_m(x)$ which has the form

$$f_{N,m}(x) \simeq g_m(x) = \frac{1}{4\pi} \frac{3}{2^m} P_1(x) = \frac{1}{4\pi} \frac{3}{2^m} x.$$

During first trials using $m = 3$ and $m = 4$, the interpolation-matrix

$$A_{\Xi} = f_{N,m}(\cos)$$

proved to be ill-conditioned as well, leading to the typical problems when trying to solve $v = A_{\Xi}b$, even by means of a least-squares solution. This is not too surprising, since the dependence on the cosine matrix is nearly linear for those values of m .

The use of $m = 2$ led to good results by using $N > 15$ terms, but since the convergence of the sum $f_{N,m}$ is slower for $m = 2$ and the error decreased in N we decided to choose $N = 40$ to be used in further investigation.

2.4. Methods used for evaluating the quality of interpolation

The data investigated derived from 30-channel full-scalp EEG measurements of 10 normal subjects, randomly chosen from data recorded in the course of a doctoral dissertation (Würzer, 2013). The EEG was amplified with a Braintronics ISO 1032 amplifier and digitised at a rate of 500 Hz, with the time-constant for the input set to 0.3 s (equivalent to highpass-filtering with a cut-off frequency of 0.53 Hz), a notch filter rejecting mains hum at 50 Hz, and a high cut-off of 70 Hz at 24 db per octave. All of the subjects were recorded in 4 situations. This way, 40 samples free

of technical artefacts of length 5 s were chosen (2501 points in time), hence every of the 40 measurements included 72529 data points. We evaluated the methods by using a leave-one-out cross-validation. The value at each site was predicted using the information of the remaining electrode sites and compared to the actual value. The error was determined as

$$MSE = \frac{1}{n} \sum_{i=1}^n (\tilde{Y}_i - Y_i)^2,$$

where $\tilde{Y}$ denotes the measured data and Y denotes the predicted data and n is the number of data sites to be reconstructed.

2.5. Results

To evaluate the methods we reconstructed the data of one electrode using the information of the other electrodes. Fig. 4 shows the result for the same data set once reconstructed using the nearest-neighbour interpolation and once reconstructed with a radial basis function method, using the multiquadric basis function and smoothing parameter $c = 0.07$.

The mean of the errors determined by the 40 datasets was computed for better comparison. The result is shown in Fig. 5. Additionally, the maximum error measured during the leave-one-out cross-validation is shown in Table 1a. One can see that the best results in both error measures were achieved using the Euclidean multiquadric method.

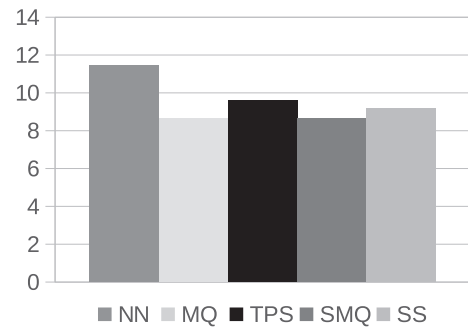
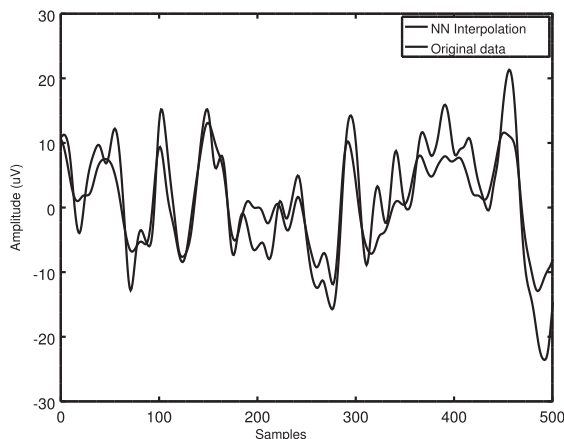
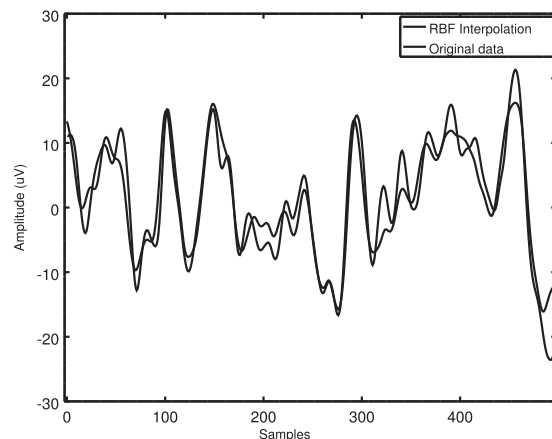


Fig. 5. The mean square errors given by the leave-one-out crossvalidation for nearest-neighbour – (NN), multiquadric – (MQ), thin-plate-spline – (TPS), spherical-multiquadric – (SMQ) and spherical spline – (SS) interpolation.



(a) Nearest-neighbour interpolation.



(b) Multiquadric interpolation.

Fig. 4. Example for a reconstruction of one electrode's data using nearest-neighbour (a) and multiquadric (b) interpolation.

Table 1

Average mean square errors for both types of crossvalidation and nearest-neighbour – (NN), multiquadric – (MQ), thin-plate-spline – (TPS), spherical-multiquadric – (SMQ) and spherical spline – (SS) interpolation.

(a) Leave-one-out-crossvalidation					
	NN	MQ	TPS	SMQ	SS
MSE	11.442 μV^2	8.651 μV^2	9.576 μV^2	8.867 μV^2	9.174 μV^2
Max	48.8 μV	44.46 μV	47.7 μV	44.4 μV	46.07 μV
(b) Leave-two-out-crossvalidation					
	MQ	TPS	SMQ	SS	
MSE	8.794 μV^2	9.761 μV^2	8.794 μV^2	9.043 μV^2	
Max	44.7 μV	48.6 μV	44.7 μV	44.6 μV	

Table 2

Evaluation of the differences between the different interpolation methods (nearest-neighbour (NN), multiquadric (MQ), thin-plate-spline (TPS), spherical-multiquadric (SMQ), and spherical spline (SS)) according to a sign-test of the mean-square-error. Bonferroni-corrected significance levels: $\diamond : \leq 10^{-4}$, $\heartsuit : \leq 10^{-7}$, $\spadesuit : \leq 10^{-10}$.

↓ better than →	NN	MQ	TPS	SMQ	SS
NN		$\diamond$			
MQ	$\spadesuit$		$\spadesuit$		$\heartsuit$
TPS	$\diamond$	$\diamond$		$\diamond$	
SMQ	$\spadesuit$		$\spadesuit$		$\heartsuit$
SS	$\diamond$		$\spadesuit$	$\diamond$	

We also investigated the errors by a leave-two-out-cross-validation, where data of two electrodes were determined from the rest. We did not include the results of the nearest-neighbour interpolant in this test, because for some electrode combinations (a corner electrode and one of its two neighbours), the data of the corner electrode would not be reconstructed, but just *replaced* by the data of its only remaining neighbour. The results are shown in Table 1b. In this test setting, the methods MQ and SMQ produced the smallest mean square error, but the spherical splines minimised the maximum error.

To test the significance of the results, we used a sign test on the data from the 40 test segments. For every pair of methods we determined the difference between the computed MSE and then used a sign test to decide whether the observed distribution of signs was significant. The results are given in Table 2, and they show that even in the worst case, occurring in the comparison of the two multiquadric-methods to spherical splines, the multiquadrics perform significantly better.

3. Discussion and conclusion

In this work, the predictive value of an interpolation technique not yet used in EEG reconstruction, the multiquadric method, was compared to commonly used interpolation techniques.

To do this, 40 samples from 10 subjects were investigated by cross-validation. Each technique was used to determine the error in predicting the potential at one or two electrode sites from the potentials recorded at the others.

Our results show that the global techniques perform better than the local techniques. This matches the results of Soufflet et al. (1991), where the performance of interpolation techniques was compared. The techniques included thin-plate-splines and spherical splines, but not the multiquadric methods. Comparing the multiquadric methods to the nearest-neighbour technique we see a significant decrease of the error. We also compared the methods using an array of 64 electrodes (a slight modification of the IFCN

10/10 System (Nuwer et al., 1998; Jurcak et al., 2007)), and could confirm that in this context, the multiquadric also performs better than the other methods, and this also showed that radial basis functions can be easily adopted to different electrode distributions. For the sake of brevity, however, we do not discuss those other results here, as they provide only little further information.

Testing the pairwise significance of the results shown in Table 1 showed that multiquadric methods performed invariably better than any other method we compared them to. In particular, while the difference in mean-square-error is about 0.3 μV^2 between spherical multiquadrics and spherical splines, it is almost 3 μV^2 when compared to nearest-neighbour interpolation, amounting to about 30% of the mean-square-error of the latter.

We conclude that in this small study the multiquadric methods performed better than the thin-plate-splines and spherical splines methods. Regarding the two multiquadric techniques, there was no significant difference in the use of Euclidean or spherical geometry. This result was also obtained by Soufflet et al. (1991). For all these reasons, we decided on the multiquadric interpolation as best suited for reconstructing EEG data. It showed good results and can be calculated easily, while outperforming nearest-neighbour interpolation by a large amount, and spline methods to a somewhat lesser extent.

Conflicts of interest

None of the authors have potential conflicts of interest to be disclosed.

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